NEUROMORPHIC COMPUTING — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 31, 2026

Diagram-based questions on Neuromorphic Computing

Question 1

10.0 points

Examine the diagram below of a leaky integrate-and-fire (LIF) neuron model, which shows the membrane potential trace over time. Analyze how the membrane potential changes in response to incoming spikes and explain the conditions under which the neuron fires. Specify any key parameters illustrated in the diagram, such as threshold potential and leakage rate.

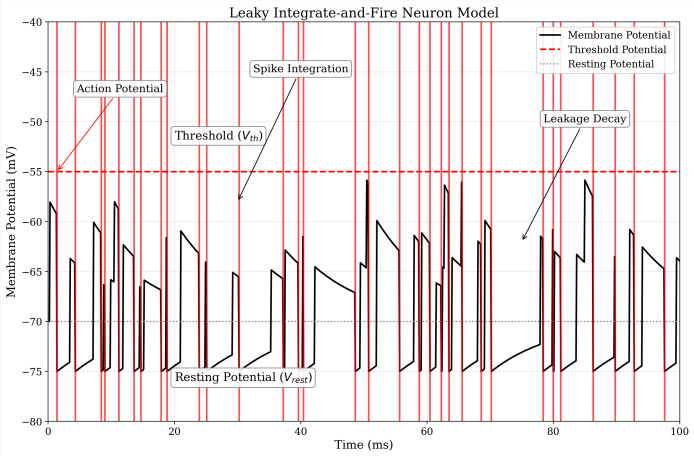

Fig. 1. Diagram of a leaky integrate-and-fire (LIF) neuron model showing the membrane potential trace over time. The diagram includes labeled axes for time and membrane potential, incoming spikes, threshold potential, and leakage rate. The membrane potential trace should show how it changes in response to incoming spikes and when it crosses the threshold potential, indicating neuron firing. IMPORTANT: The diagram is missing required labels. The diagram description requires. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram is missing required labels for "incoming. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 2 10.0 points

Examine the diagram below illustrating a spike-timing dependent plasticity (STDP) learning rule. Describe how the timing of pre- and postsynaptic spikes affects the synaptic weight change. Discuss the implications of synaptic modifications based on different time intervals between spikes.

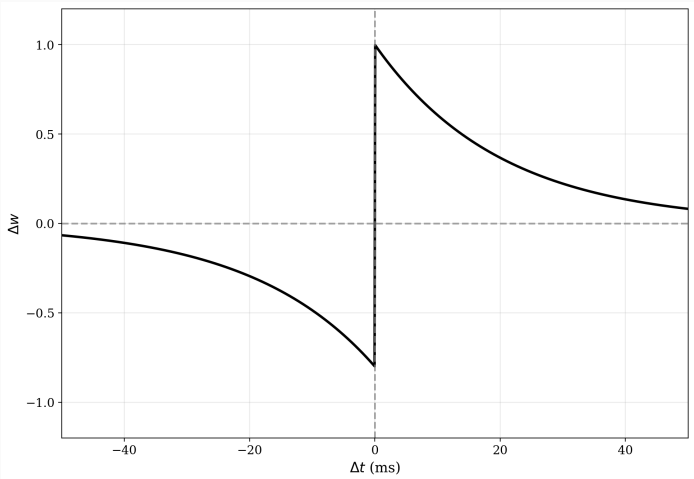

Fig. 2. A typical STDP curve showing the relationship between the time difference (Δt) between pre- and postsynaptic spikes and the change in synaptic weight (Δw). The x-axis represents Δt, with positive values indicating pre-before-post and negative values indicating post-before-pre. The y-axis represents Δw, showing potentiation for positive Δt and depression for negative Δt. IMPORTANT correction needed: The diagram contains labels that explicitly describe the phenomena the student. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 3

10.0 points

Examine the diagram below of a crossbar array architecture used for storing synaptic weights in neuromorphic computing. Explain how this architecture enables efficient weight storage and retrieval. Highlight any challenges related to scaling and variability illustrated in the diagram.

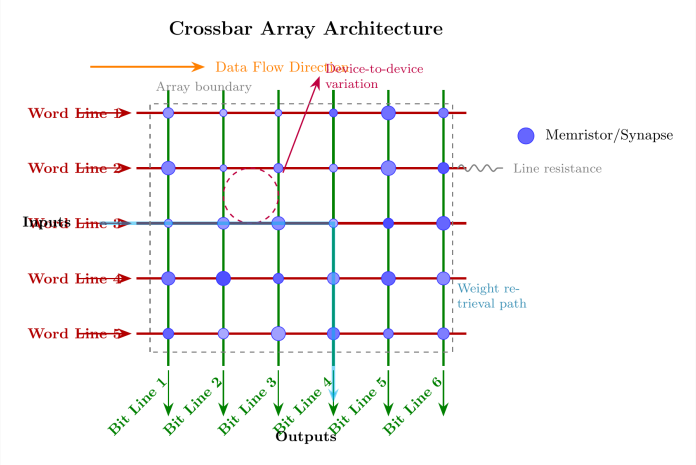

Fig. 3. A crossbar array architecture for neuromorphic computing. The diagram should show a grid of intersecting horizontal and vertical lines representing the crossbar structure. At each intersection, there should be a symbol representing a synapse or memory cell, indicating where synaptic weights are stored. The diagram should also include labels for input lines (rows) and output lines (columns), and highlight potential variability issues such as resistance variations at intersections. Additionally, include annotations to indicate challenges related to scaling, such as increased line resistance and signal degradation in larger arrays. IMPORTANT correction needed: The diagram contains multiple annotations that directly explain the concepts the. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT: The diagram is missing required labels. The diagram fails to include the required annotations illustrating the challenges. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 4

10.0 points

Evaluate the temporal coding scheme in the spiking neural network shown in the diagram below. Describe how information is encoded temporally and its impact on network processing capabilities. What are the advantages of temporal coding compared to rate coding based on the diagram?

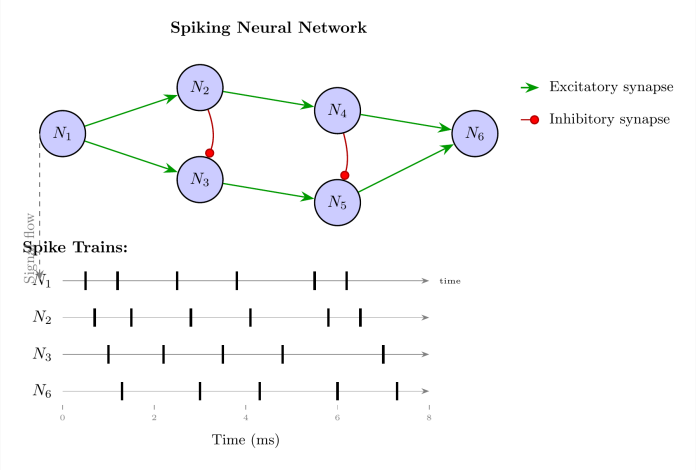

Fig. 4. CUMULATIVE LEAK CORRECTIONS (all must be fixed): The "Key Features" box explicitly lists the core; The diagram contains labels "Variable inter-spike intervals. A spiking neural network diagram showing neurons and synapses. Include several neurons connected by synapses. Illustrate temporal coding by showing spike trains with different timing patterns. Label neurons and synapses, and indicate the direction of signal flow. Highlight the temporal aspect by using time markers or arrows to show the sequence of spikes. Compare temporal coding with rate coding by showing a separate rate coding example with average firing rates. IMPORTANT correction needed: The "Key Features" box explicitly lists the core. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT correction needed: The diagram contains labels "Variable inter-spike intervals. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 5 10.0 points

Examine the comparative diagram of energy consumption between von Neumann and neuromorphic architectures shown in the diagram below. Discuss the reasons for differences in energy efficiency. What implications does this have for large-scale neural applications?

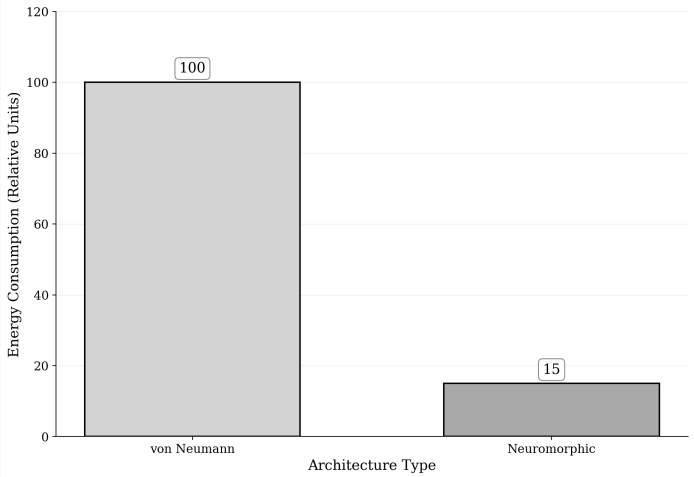

Fig. 5. A comparative energy consumption chart showing the differences between von Neumann and neuromorphic architectures. The chart should have two bars or lines representing each architecture, with energy consumption values on the y-axis and architecture types on the x-axis. This will help visualize the energy efficiency differences between the two architectures.